

CSMVessel0000000005 Entertainment V2.0

Project Noah's Aegis: Maritime Entertainment Systems Resilience Under High-Flux EM Events

Version 2.0 | June 2026

Changes V1.0 to V2.0

- GaN solid-state audio amplifiers: 97% efficiency (same tech as Safe-Wave CSMFAB005) applied
- LED theatrical lighting: all-ceramic driver ICs in BFRP housings (GIC-immune)
- OLED display panels: flexible BFRP-backed OLED for structural compliance
- MXene-shielded AV server rooms: SE=92 dB (replaces V1.0 copper enclosure SE=45 dB)
- Acoustic design: Silence-Block (CSMFAB032) LRAM technology scales to theater scale
- Solar Cycle 25: entertainment systems = non-safety IT -> lowest priority in failure hierarchy

1. Entertainment System Vulnerability V2.0

Cruise ship entertainment systems typically include: - Main theater: 1,200+ seats, full stage rigging, lighting, audio (\$8-15M value) - Casino: ~500 slot machines, 40+ gaming tables (electronic + conductive) - Multiple restaurants: AV systems, automated beverage systems - Lido deck: outdoor LED screens, weather-proofed audio - Fiber network: ship-wide WiFi, IPTV

V2.0 GIC risk assessment per system:

System	GIC Risk	Failure Mode	Priority
Casino slot machines	CRITICAL	PCB burnout at 43 V/km	Revenue critical
Main theater lighting (LED drivers)	HIGH	Driver IC destruction	Operations
Main theater audio (copper amps)	HIGH	Output stage burnout	Operations
IPTV/WiFi infrastructure	HIGH	Switch/router burnout	Passenger exp.
Navigation PA / emergency broadcast	CRITICAL	Must survive -> Aegis priority	Safety

2. V2.0 Aegis Entertainment Systems

2.1 GaN Audio Amplifiers (per CSMFAB00000000005 Science)

Same GaN HEMT technology as Safe-Wave microwave source: - Drain efficiency: 97% (vs Class AB 65%) - V2.0: all audio amplifiers specified as GaN Class-D or Class-F topology - GaN chip packaging: MXene-shielded ceramic package (SE=92 dB for GHz susceptibility) - Signal input: PMMA optical fiber from DSP (zero metal signal path)

Ship-wide audio system GIC immunity: - All copper signal cables replaced with PMMA POF (per CSMFAB000000000013 design) - Power amplifiers: GaN in BFRP-cased, MXene-shielded enclosures - Loud-speaker wiring: retained copper (OK for power wiring with DC-blocking capacitors) - Result: zero GIC signal corruption; amplifiers survive indefinitely in storm field

2.2 LRAM Theater Acoustic Insulation

MIT 2025 LRAM technology (per CSMFAB000000000032 Silence-Block): Scaled from automotive (4-door panels) to theater scale (60m2 stage wall): - WC sphere resonators (d=8.5mm) in magnetite-doped PDMS

matrix - Panel: 15mm thick, -65 dB at 250 Hz - Theater separation: adjacent theater sound bleed at 250 Hz reduced 99.97% - Ship vibration: main-theater-to-seating vibration at 25 Hz: -31 dB reduction

2.3 Casino Slot Machine GIC Protection

Each slot machine = a sealed inductive loop (power cord + internal PCB copper) At 43 V/km across casino floor (30m): $V_{\text{induced}} \sim 1.3 \text{ V}$; I in low-R PCB traces = significant currents.

V2.0 solution per machine: - BFRP outer case (replaces ABS plastic + steel frame): \$180/machine conversion - MXene inner shield: \$45/machine - Optical fiber payment terminal: \$220/machine Total per machine: \$445; 500 machines: \$222,500/ship vs. one storm event destroying all machines: \$5M+ replacement cost.

End of CSMVessel0000000005 Entertainment V2.0 | Carrington Storm Motors